

Assignment-2

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Q.Create an Account class Heirarchy

Account with super class (acc_id, name, balance)

methods - withdraw and deposit

Create SavingsAccount as sub class of account - additional field (type - personal/corporate etc)

implement withdraw and deposit such that

- maximum upto 1 lakh can be deposited in an account at a time
- Min balance 5000 must be maintained while withdrawal (if type = corporate you withdraw full amount = balance)

Create CurrentAccount as sub class of account

implement withdraw and deposit such that

- maximum upto 2 lakh can be deposited in an account at a time
- Min balance 10000 must be maintained while withdrawal

Create Bank App with Transaction class

Create Method withdraw_from_account(account : Account) and deposit_to_account(account : Account)

These methods will return the new balance after deposite/withdraw

Create user class with user interface that gives 2 menu options

1. Deposit
2. Withdraw

Both options will ask user to enter money to withdraw/deposite

Display a statement with each transaction and final balance after user exits from the menu

Identify possible Exceptions and implement them

Program:-

Account.py

```
from abc import ABC, abstractmethod
from Exceptions import DepositLimitError, MinimumBalanceError,
InsufficientBalanceError

class Account(ABC):
    def __init__(self, acc_id, name, balance):
```

```

        self.acc_id=acc_id
        self.name=name
        self.balance=balance
        self.transactions=[]

    def __str__(self):
        return f'{self.acc_id} {self.name} {self.balance}'

    def get_balance(self):
        return self.balance

    @abstractmethod
    def withdraw(self, amount):
        pass

    @abstractmethod
    def deposit(self, amount):
        pass

class SavingsAccount(Account):
    def __init__(self, acc_id, name, balance, account_type):
        super().__init__(acc_id, name, balance)
        self._account_type=account_type

    # def __str__(self):
    #     return f'{self._acc_id} {self._name} {self._balance}
    {self._account_type}'

    def withdraw(self, amount):
        if self._account_type=='corporate':
            if self.balance-amount>=5000:
                self.balance -=amount
            else:
                raise MinimumBalanceError(f'Min balance 5000
must be maintained.....')
        else:
            if self.balance-amount<0:
                raise InsufficientBalanceError('Insufficient
balance')
            else:
                self.balance -= amount
                self.transactions.append(f'Withdrawn: {amount} Balance:
{self.balance}')
        return self.balance

```

```

def deposit(self, amount):
    if amount <= 100000:
        self.balance += amount
        self.transactions.append(f'Deposited: {amount}
Balance: {self.balance}')
    else:
        raise DepositLimitError('Maximum upto 1 lakh can be
deposited in an account at a time')
    return self.balance

class CurrentAccount(Account):
    def __init__(self, acc_id, name, balance):
        super().__init__(acc_id, name, balance)

    # def __str__(self):
    #     return f'{self._acc_id} {self._name} {self._balance}'

    def withdraw(self, amount):
        if self.balance - amount >= 10000:
            self.balance -= amount
            self.transactions.append(f'Withdrawn: {amount}
Balance: {self.balance}')
        else:
            raise MinimumBalanceError(f'Min balance 10000 must
be maintained while withdrawal')
        return self.balance

    def deposit(self, amount):
        if amount <= 200000:
            self.balance += amount
            self.transactions.append(f'Deposited: {amount}
Balance: {self.balance}')
        else:
            raise DepositLimitError('Maximum upto 2 lakh can be
deposited in an account at a time')
        return self.balance

```

Transaction.py

```
class Transaction:

    @staticmethod
    def withdraw_from_account(account, amount):
        balance = account.withdraw(amount)
        return balance

    @staticmethod
    def deposit_from_account(account, amount):
        balance = account.deposit(amount)
        return balance
```

Exceptions.py

```
class DepositLimitError(Exception):
    pass

class MinimumBalanceError(Exception):
    pass

class InsufficientBalanceError(Exception):
    pass

class InvalidAccountNumberError(Exception):
    pass
```

User.py

```
from Account import SavingsAccount, CurrentAccount
from Transaction import Transaction
from Exceptions import *

accounts = {
    101: CurrentAccount(101, 'Ashlesha', 60000),
    102: CurrentAccount(102, 'Sakshi', 40000),
    103: SavingsAccount(103, 'Akash', 40000, 'personal'),
    104: SavingsAccount(104, 'Prasad', 7000, 'corporate'),
    105: SavingsAccount(105, 'Pratik', 60000, 'personal')
}

try:
```

```

acc_no=int(input("Enter Account Number: "))
if acc_no not in accounts:
    raise InvalidAccountNumberError("Invalid Account
Number")

acc_list=accounts[acc_no]
t=Transaction()
flag=True

while flag:
    print("\n---BANK ACCOUNT---")
    print("1. Deposit")
    print("2. Withdraw")
    print("3. Exit")

    # amount=int(input('Enter amount which you want to
withdraw or deposit: '))
    choice = int(input("\nSelect an option (1-3): "))
    try:
        match choice:
            case 1:
                amount = int(input('Enter amount to
deposit: '))

                bal=t.deposit_from_account(acc_list,amount)
                print('Deposit Successful.....')
                print("Current Balance: ",bal)
            case 2:
                amount = int(input('Enter amount to
withdraw: '))

                bal=t.withdraw_from_account(acc_list,amount)
                print('Withdraw Successful.....')
                print("Current Balance: ",bal)
            case 3:
                flag=False
                print('Transaction Statement')
                print('Account Number: ',acc_list.acc_id)
                print('Name: ',acc_list.name)
                if len(acc_list.transactions)==0:
                    print("No transactions done")
                else:
                    for i in acc_list.transactions:
                        print(i)

```

```
        print("Final Balance:
",acc_list.get_balance())
        print('Thank you for Banking!!!')
    case _:
        print("Invalid option. Please choose 1, 2,
or 3.")

    except DepositLimitError as e:
        print("Error:", e)

    except MinimumBalanceError as e:
        print("Error:", e)

    except InsufficientBalanceError as e:
        print("Error:", e)

    except Exception as e:
        print("Unexpected Error:", e)

except InvalidAccountNumberError as e:
    print("Error: ",e)
except ValueError:
    print("Invalid input! Please enter numbers only.")
```

Outputs:-

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day6\Assignme
Enter Account Number: 101
```

```
---BANK ACCOUNT---
```

1. Deposit
2. Withdraw
3. Exit

```
Select an option (1-3): 1
```

```
Enter amount to deposit: 5000
```

```
Deposit Successful.....
```

```
Current Balance: 65000
```

```
---BANK ACCOUNT---
```

1. Deposit
2. Withdraw
3. Exit

```
Select an option (1-3): 2
```

```
Enter amount to withdraw: 4000
```

```
Withdraw Successful.....
```

```
Current Balance: 61000
```

```
Current Balance: 61000
```

```
---BANK ACCOUNT---
```

1. Deposit
2. Withdraw
3. Exit

```
Select an option (1-3): 1
```

```
Enter amount to deposit: 300000
```

```
Error: Maximum upto 2 lakh can be deposited in an account at a time
```

```
---BANK ACCOUNT---
```

1. Deposit
2. Withdraw
3. Exit

```
Select an option (1-3): 2
```

```
Enter amount to withdraw: 52000
```

```
Error: Min balance 10000 must be maintained while withdrawal
```

```
---BANK ACCOUNT---
```

1. Deposit

```
---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 3
Transaction Statement
Account Number: 101
Name: Ashlesha
Deposited: 5000 Balance: 65000
Withdrawn: 4000 Balance: 61000
Final Balance: 61000
Thank you for Banking!!!

Process finished with exit code 0
```

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day6\Assign
Enter Account Number: 103

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 1
Enter amount to deposit: 60000
Deposit Successful.....
Current Balance: 100000

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 1
Enter amount to deposit: 150000
Error: Maximum upto 1 lakh can be deposited in an account at a time
```

```
---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 2
Enter amount to withdraw: 6000
Withdraw Successful.....
Current Balance: 94000

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 2
Enter amount to withdraw: 95000
Error: Insufficient balance

---BANK ACCOUNT---
1. Deposit
```



```
---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 3
Transaction Statement
Account Number: 103
Name: Akash
Deposited: 60000 Balance: 100000
Withdrawn: 6000 Balance: 94000
Final Balance: 94000
Thank you for Banking!!!

Process finished with exit code 0
|
```

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day6\Assi
Enter Account Number: 104

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 1
Enter amount to deposit: 5000
Deposit Successful.....
Current Balance: 12000

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 1
Enter amount to deposit: 150000
Error: Maximum upto 1 lakh can be deposited in an account at a time
```

```

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 2
Enter amount to withdraw: 1000
Withdraw Successful.....
Current Balance: 11000

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 2
Enter amount to withdraw: 7000
Error: Min balance 5000 must be maintained.....

---BANK ACCOUNT---

```

```

3. Exit

Select an option (1-3): 2
Enter amount to withdraw: 15000
Error: Min balance 5000 must be maintained.....

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 3
Transaction Statement
Account Number: 104
Name: Prasad
Deposited: 5000 Balance: 12000
Withdrawn: 1000 Balance: 11000
Final Balance: 11000
Thank you for Banking!!!

Process finished with exit code 0
|

```

```

C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day6\Assig
Enter Account Number: gsgkhkj
Invalid input! Please enter numbers only.

Process finished with exit code 0
|

```

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day6\Assig
Enter Account Number: 102

---BANK ACCOUNT---
1. Deposit
2. Withdraw
3. Exit

Select an option (1-3): 3
Transaction Statement
Account Number: 102
Name: Sakshi
No transactions done
Final Balance: 40000
Thank you for Banking!!!

Process finished with exit code 0
|
```

```
C:\Users\PGCP-AI\AppData\Local\Python\pythoncore-3.14-64\python.exe D:\AshleshaRuchika\PGCP-AI\Python\Day6\Assig
Enter Account Number: 501
Error: Invalid Account Number

Process finished with exit code 0
```